



## Preliminary characterization of Alpine snow using SnowMicroPen

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### ABSTRACT

Objective and accurate observations of snowpack layering and properties are essential for avalanche warning. Human observations are helpful, but they are often highly subjective and therefore inconsistent making them difficult to interpret. In this paper we address this problem by applying a high-resolution penetrometer to determine snowpack properties. We develop an algorithm to characterize the different snow microstructure classes using the signal obtained from the penetrometer measurements. For this purpose a database consisting of various snow profiles of Alpine snow was created, which contains the records of the mean, standard deviation and coefficient of variation of penetration resistance over 1 mm depth. The algorithm is able to characterize the major snow classes, namely, new snow, faceted snow, depth hoar, rounded grains and melt-freeze with an acceptable accuracy for warning purposes. We find that new snow, depth hoar and melt-freeze snow layers are characterized better than the rounded grains and faceted snow layers. However, knowledge-based rules, formulated from the experience of field researchers, can further improve the automated snow profiling procedure. For a comprehensive prediction of snow classes with better accuracy, it is proposed in future to include more expert rules and to enlarge the database of measurements. This preliminary approach could be helpful in validating numerical simulation models of snowpacks and subsequently numerical avalanche forecasting. The potential application of snow class information derived from the SMP to infer the specific surface area of snow layers is also discussed.

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### 1. Introduction

The mountain snow cover is very complex and difficult to assess objectively by human observers (Sturm et al., 1996; Pielmeier and Schneebeli, 2003a). Since an Alpine snowpack can have a maximum depth of 250 cm and contain more than 15 snow layers (Sturm et al., 1995) human observations are time-consuming and often inconsistent. Salient features of the snow cover are often missed because thin snow layers cannot be objectively identified. Although melt features are generally not significant in alpine snowcovers, all important basal depth hoar or wind affected layers are usually present. The observations and hence the resolution of snow profiles, however, depends strongly on the momentarily interesting and prominent features identified by the observer. Due to the high degree of spatial variability an objective classification of mountain snow is therefore extremely difficult. Of course, the difficulty of interpreting the human observations increases when the observers are not intensively trained, establishing some kind of common standard.

Schneebeli and Johnson (1998) developed a high-resolution (4  $\mu$ m spatial) and high-speed (20 mm/s) penetrometer called SnowMicroPen (SMP), which is used to take measurement of snow-depth and snow-resistance. The SMP can detect very thin snow layers that are not picked up by hand profiles (Pielmeier and Schneebeli, 2003b). The small diameter measuring tip (5 mm) of the SMP ensures that snow resistance is measured at the tip as it penetrates the snow; the resisting force is primarily created by breaking the interconnections between snow grains. This assumption is not completely correct, but results in minimal errors for very porous materials like snow (Johnson and Schneebeli, 1999). In comparison, the larger diameter cone (40 mm) of the Swiss ramsonde (Bader et al., 1954) creates considerable compression of the snow below the probe's tip and hence it is not a true representative of the layer strength. Assuming the deformation of snow to be at the penetrometer tip, Johnson and Schneebeli (1999) developed a micro-mechanical model for snow, which is able to estimate physical and mechanical properties of snow from the SMP profile. The model calculates the number of bonds fractured during SMP penetration. However, the model is somewhat limited in that the threshold value of penetration force required to break a bond is independent of snow element size (Schneebeli, 1999). The micro-mechanical model was later used by Kronholm et al. (2004) to estimate snow properties in comparison to stuff block test scores. Pielmeier and Schneebeli (2003b) tested the quality of different hardness measurement methods

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**Table 1**  
Different snow classes introduced in the database

| Profile no.          | Date, location code | Different snow classes introduced in the database |                            |                              |                        |                     |                           |                  |
|----------------------|---------------------|---------------------------------------------------|----------------------------|------------------------------|------------------------|---------------------|---------------------------|------------------|
|                      |                     | New snow<br>(1d, 1e, 2a)                          | Rounded grains<br>(3a, 3b) | Faceted snow<br>(4a, 4b, 4c) | Depth hoar<br>(5a, 5b) | Melt-freeze<br>(6b) | Hard melt- Freeze<br>(9e) | Wet snow<br>(6a) |
| 1                    | 2003, 14Feb, WF     |                                                   |                            | 3 (183)                      |                        | 6 (24)              |                           |                  |
| 2                    | 28Feb, WF           |                                                   |                            | 3 (156)                      |                        |                     |                           |                  |
| 3                    | 31Mar, WF           |                                                   |                            |                              |                        | 3 (64)              |                           |                  |
| 4                    | 01Apr, WF-Jgipfeli  |                                                   |                            |                              |                        | 1 (150)             | 1 (7)                     |                  |
| 5                    | 16May, WF           |                                                   |                            |                              |                        | 2 (290)             |                           | 2 (460)          |
| 6                    | 25Nov, Jatz- horn   |                                                   |                            | 2 (242)                      |                        | 1 (6)               |                           |                  |
| 7                    | 29Nov, WF           | 3 (650)                                           | 3 (455)                    | 8 (715)                      |                        |                     | 1 (9)                     |                  |
| 8                    | 13Dec, WF           |                                                   | 3 (390)                    | 6 (815)                      |                        |                     |                           |                  |
| 9                    | 2004, 02Jan, WF     |                                                   | 3 (315)                    | 6 (765)                      | 3 (185)                |                     |                           |                  |
| 10                   | 16Jan, WF           |                                                   |                            | 8 (740)                      |                        | 1 (3)               |                           |                  |
| 11                   | 30Jan, WF           | 3 (1080)                                          | 3 (280)                    | 3 (335)                      |                        |                     |                           |                  |
| 12                   | 31Mar, WF           |                                                   | 1 (650)                    |                              | 5 (398)                |                     |                           |                  |
| 13                   | 22Dec, Davos        | 3 (700)                                           | 3 (900)                    | 6 (500)                      | 3 (320)                |                     |                           |                  |
| 14                   | 2005, 02Feb, Davos  | 2 (275)                                           |                            | 3 (82)                       | 2 (105)                |                     |                           |                  |
| Total no. of layers  |                     | 11                                                | 16                         | 48                           | 13                     | 14                  | 2                         | 2                |
| Total thickness (mm) |                     | 2705                                              | 2990                       | 4533                         | 1008                   | 537                 | 16                        | 460              |
| % thickness          |                     | 22.1                                              | 24.4                       | 37.0                         | 8.2                    | 4.4                 | 0.1                       | 3.8              |

ICSSG classification (Colbeck et al., 1990) is also shown.

The first value against each date shows the number of snow layers (same class) introduced from different SMP profiles and second value in the parenthesis shows the total thickness (in mm). Profiles nos. 7, 9 and 12 are used for validation. WF – Weissfluhjoch study plot.

in comparison to planar sections. While the SMP recorded all the identified layers and boundaries, classical hand profiles only reflect 80% thereof. Based on this information, Satyawali et al. (2004) used SMP profiles to develop a database for a Himalayan snowpack and then estimated snow layers. The approach is found useful in finding out the snow type (snow class, which is mainly characterized from its texture) according to the international snow classification (Colbeck et al., 1990), but the accuracy of their approach could not be established. They also compared penetration force data of SMP with the snow resistance data of the ramsonde, and found that a direct comparison is not possible. With this background knowledge, we are convinced that objective snow characterization is an important step before estimating the micro-mechanical properties from the SMP profile.

In the present work, a similar approach to Satyawali et al. (2004) is used to characterize snow with the help of a database developed for Alpine snow from the SMP profiles of three winters. The database is prepared from the penetration resistance data of different snowpacks acquired through SMP measurements. The SMP profiles taken at the SLF-study plot Weissfluhjoch (2540 m) and near Davos (1560 m), Switzerland are used in the database. Finally, an attempt is made to characterize snow from SMP profiles. The analysis shows that five major snow classes, which are new snow, faceted snow, depth hoar, rounded grains and melt-

freeze snow can be characterized well. The database helped in picking up the similar signals in SMP profiles, and it further required implementing knowledge based rules in identifying and positioning a snow layer in a snow profile. Expert rules are used because it is recognized that the characteristics of snow layers and the sequence of layers comprising a snowpack arise from the known winter climate. Therefore, this approach is only applicable to snowpacks in the Alpine climate zone. The success of this approach depends on the database size and the applied rules. The accuracy of this algorithm is also estimated in the present work.

This work is an effort towards an automatic and observer-independent snow classification. It is assumed that there is a certain class of snow in a particular snow layer but often transitions are found because snow is changing continuously, leading to transitions from one class to another. The international snow classification scheme (Colbeck et al., 1990) and classical snow stratigraphy are not enough to characterize such transitions. The present numerical scheme intends to retrieve and classify alpine snow properties. Classification of snow is often used to help assess the current and future stability of the snowpack for avalanche forecasting. While the presented algorithm is a helpful step toward the classification of snow, the SMP signal can

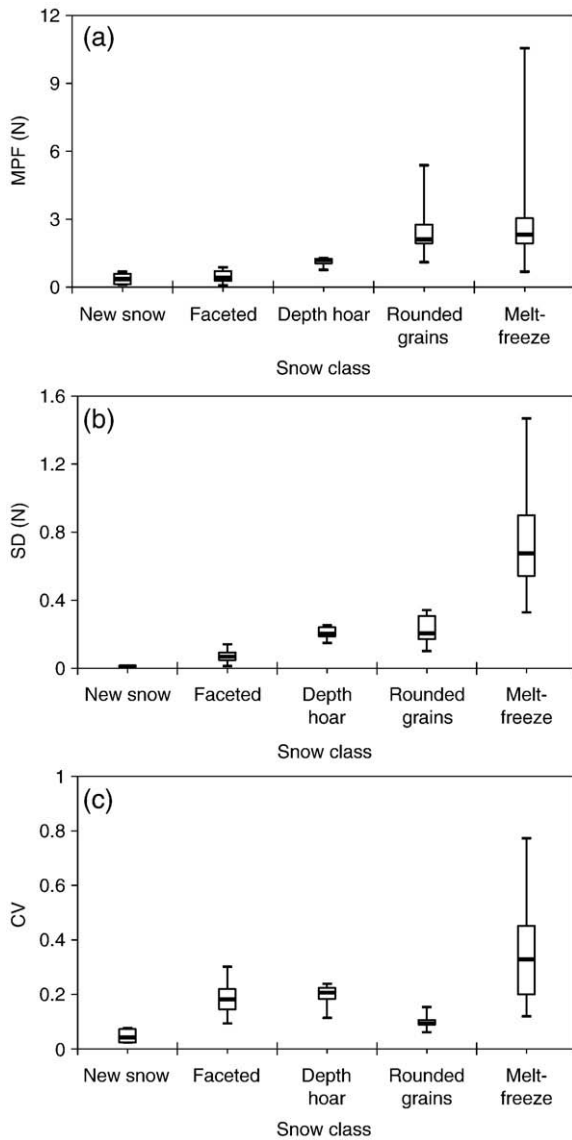
**Table 2**  
Different variables with their minimum, maximum, mean and standard deviation values extracted from the database for five major snow classes

| Snow classes / ICSSG code (Colbeck et al., 1990) | Variables | Minimum | Maximum | Mean  | Standard deviation | Total no. of records used in database |
|--------------------------------------------------|-----------|---------|---------|-------|--------------------|---------------------------------------|
| New snow/1d, 1e, 2a                              | MPF (N)   | 0.008   | 0.873   | 0.286 | 0.280              | 2705                                  |
|                                                  | SD (N)    | 0.003   | 0.039   | 0.010 | 0.006              |                                       |
|                                                  | CV        | 0.007   | 0.740   | 0.111 | 0.113              |                                       |
| Rounded grains/ 3a, 3b                           | MPF (N)   | 0.039   | 10.3    | 2.277 | 2.216              | 2990                                  |
|                                                  | SD (N)    | 0.007   | 0.857   | 0.179 | 0.148              |                                       |
|                                                  | CV        | 0.007   | 0.353   | 0.113 | 0.052              |                                       |
| Faceted/4a, 4b, 4c                               | MPF (N)   | 0.016   | 1.55    | 0.378 | 0.264              | 4533                                  |
|                                                  | SD (N)    | 0.004   | 0.370   | 0.066 | 0.044              |                                       |
|                                                  | CV        | 0.029   | 0.649   | 0.201 | 0.087              |                                       |
| Depth hoar/ 5a, 5b                               | MPF (N)   | 0.047   | 2.428   | 0.797 | 0.551              | 1008                                  |
|                                                  | SD (N)    | 0.012   | 0.736   | 0.165 | 0.094              |                                       |
|                                                  | CV        | 0.046   | 1.207   | 0.284 | 0.162              |                                       |
| Melt-freeze and hard crust/ 6b, 9e               | MPF (N)   | 0.421   | 23.456  | 6.497 | 4.951              | 553                                   |
|                                                  | SD (N)    | 0.128   | 5.691   | 1.076 | 0.764              |                                       |
|                                                  | CV        | 0.028   | 2.015   | 0.242 | 0.176              |                                       |

**Table 3**  
Three variables shown in Table 2 at 1 mm scale are now averaged over layer thickness scale for all SMP profiles

| Snow classes/ ICSSG code (Colbeck et al., 1990) | Variables | Min    | Max    | Mean  | Standard deviation | Number of homogeneous layers used for PCA analysis |
|-------------------------------------------------|-----------|--------|--------|-------|--------------------|----------------------------------------------------|
| New snow/1d, 1e, 2a                             | MPF (N)   | 0.109  | 0.6757 | 0.369 | 0.277              | 6                                                  |
|                                                 | SD (N)    | 0.0082 | 0.016  | 0.012 | 0.004              |                                                    |
|                                                 | CV        | 0.0229 | 0.0766 | 0.047 | 0.027              |                                                    |
| Rounded grains/ 3a, 3b                          | MPF (N)   | 1.1    | 5.388  | 2.52  | 1.187              | 10                                                 |
|                                                 | SD (N)    | 0.1018 | 0.3423 | 0.229 | 0.083              |                                                    |
|                                                 | CV        | 0.0612 | 0.1534 | 0.1   | 0.031              |                                                    |
| Faceted/4a, 4b, 4c                              | MPF (N)   | 0.072  | 0.8658 | 0.451 | 0.246              | 32                                                 |
|                                                 | SD (N)    | 0.0133 | 0.141  | 0.074 | 0.038              |                                                    |
|                                                 | CV        | 0.093  | 0.3012 | 0.182 | 0.054              |                                                    |
| Depth hoar/5a, 5b                               | MPF (N)   | 0.7608 | 1.2799 | 1.113 | 0.204              | 8                                                  |
|                                                 | SD (N)    | 0.1485 | 0.2526 | 0.209 | 0.038              |                                                    |
|                                                 | CV        | 0.1137 | 0.2383 | 0.196 | 0.042              |                                                    |
| Melt-freeze and hard crust/6b, 9e               | MPF (N)   | 0.6846 | 10.55  | 3.44  | 3.281              | 12                                                 |
|                                                 | SD (N)    | 0.3292 | 1.4691 | 0.766 | 0.368              |                                                    |
|                                                 | CV        | 0.1194 | 0.7731 | 0.34  | 0.187              |                                                    |

The total data points thus reduce from 11789 1-mm-layers (Table 2) to 68 homogeneous layers. The box plot of the data is shown in Fig. 1.



**Fig. 1.** Three representative single-value discrimination box plots using (a) MPF, (b) SD and (c) CV are shown. Boxes span the interquartile range from 1st to 3rd quartile with a horizontal line showing the median. Whiskers show the range of the values in each class. Notice, no single variable is a good discriminator.

also give direct information about the gradient of penetration resistance and hence shear stress concentrations (Schweizer, 1993). The advantage of the grain shape classification includes a mechanical interpretation of the SMP signal in terms of snowpack stability, which is currently lacking. In general, such classification systems are useful in many ways. They allow a complex description of attributes or forms to be reduced into code and second they allow deducing more data about the material than is actually measured (Sturm et al., 1995).

Finally, we discuss the potential of the SnowMicroPen to estimate specific surface area (Matzl and Schneebeli, 2006). Specific surface area (SSA) is a key parameter for snow, and based on classification developed here, a reasonable estimate can be made for SSA.

## 2. Methods

### 2.1. Snow profiling

SMP profiling was done throughout the winter, covering early winter to spring snowpack conditions to acquire data of various snow

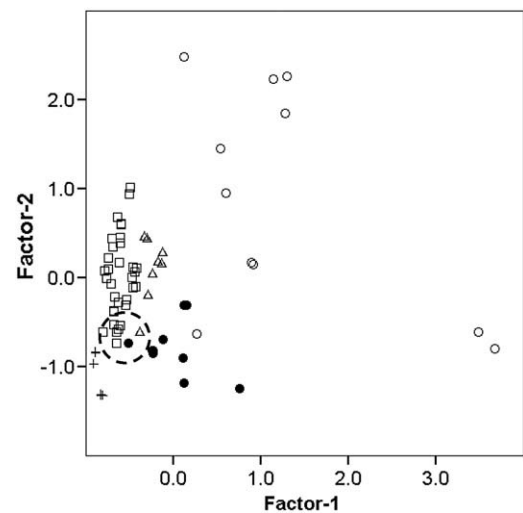
classes. The study is based on measurements made at the SLF-study plot Weissfluhjoch (2540 m) for two winters (2002–03 and 2003–04) and for one winter (2004–2005) at Davos (1560 m). Hand profiles were also taken in the vicinity (0.15–0.20 m) of SMP profiles to compare the two methods and to minimize spatial changes. The snow parameters, which were measured in the hand snow profiles are snow class, snow layer thickness, hand hardness, layer density and grain size as well as ram resistance.

### 2.2. Development of the database

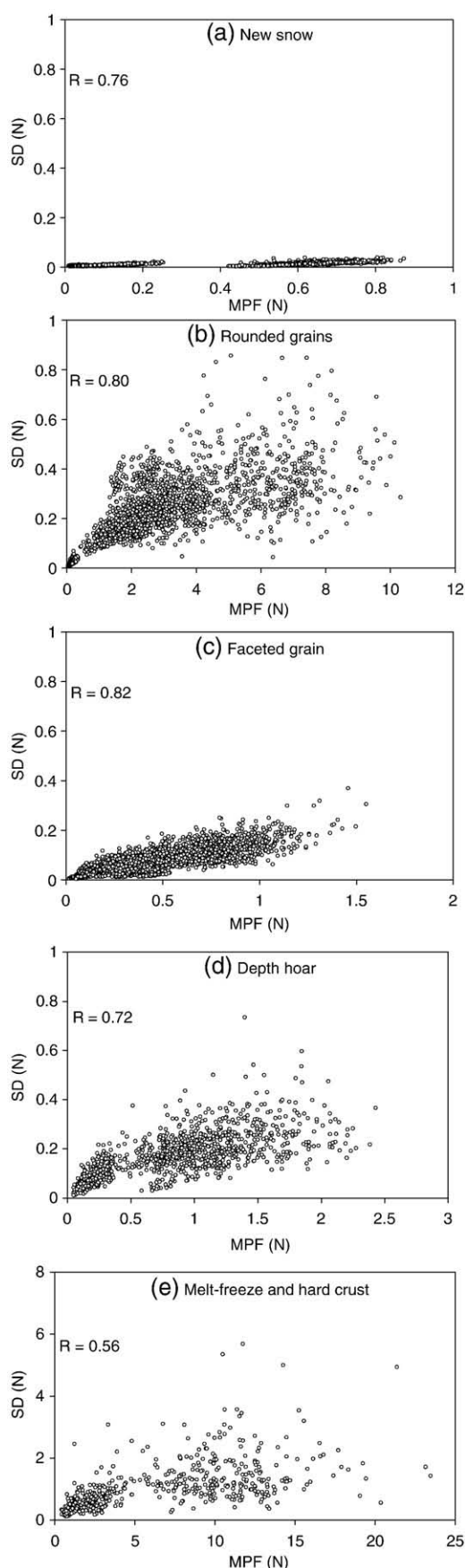
Table 1 shows the details of the SMP profiles used in the database. Only major snow classes are recorded in the database. Some layers in the hand snow profiles did not belong to a particular snow class but to a mixture of two snow classes. Such classes were not considered in the development phase of the database. The database has the highest number records for faceted snow (class 4a, 4b and 4c) and the least number of records for hard melt-freeze (class 9e). We have chosen a group of variables derived directly from the penetration force data of SMP to discriminate snow classes. Three variables used in the database are the mean of the penetration force (MPF) calculated for a 1 mm window of SMP profile (250 records in 1 mm), its standard deviation (SD) and the coefficient of variation (CV). CV is defined as the ratio of the SD to the MPF as done by Schneebeli et al. (1999). The ranges of MPF, SD and CV are obtained for five major snow classes with their mean values and standard deviation (Table 2). The highest value of mean MPF (6.5 N) is found for melt-freeze snow (class 6b, 9e) and the lowest value (0.286 N) for new snow. Mean SD also shows a similar trend as mean MPF. It decreases in the order from melt-freeze (1.076 N), rounded grains (0.179 N), depth hoar (0.165 N), faceted (0.066 N) to new snow (0.01 N). SD of depth hoar and rounded grains snow is found to be similar for MPF below 2.5 N. The highest value of mean CV (0.284) is found for depth hoar snow and lowest value (0.111) for new snow. Mean CV decreases in the order from depth hoar, melt-freeze, faceted, rounded grains to new snow. New snow and rounded grains snow is having similar values of CV below 1.0 N of MPF.

#### 2.2.1. Discriminating snow classes using average values of MPF, SD and CV for homogeneous snow layer

Values of three variables estimated in Table 2 are further averaged at layer thickness scale rather than 1 mm scale, for all homogeneous



**Fig. 2.** Principal component two-factor analysis plot based on three variables (MPF, SD and CV) is shown (ref. Table 3). Most of the snow classes are discriminated well by two-factors with maximum variance explained as 97%. Dotted circle shows mix classification for a small region. Legend: + New snow, □ Faceted snow, Δ Depth hoar, ● Rounded grains and ○ Melt-freeze.



**Fig. 3.** SD values for five major snow classes is plotted against MPF at 1 mm scale (a to e). It shows SD and MPF are positively correlated. Correlation coefficient ( $R$ ) is also calculated. Statistics of all snow classes are given in Table 2.

**Table 4**

Correlation coefficient matrix of three variables used in the database is shown

|     | MPF    | SD    | CV |
|-----|--------|-------|----|
| MPF | 1      |       |    |
| SD  | 0.881  | 1     |    |
| CV  | -0.081 | 0.223 | 1  |

snow layers. The number of data points thus reduces from 11,789 (Table 2) to 68 (Table 3). Fig. 1 shows standard box plot for five snow classes using a single variable. Vertical lines show the range of the values of each class. In each variable (i.e. MPF, SD and CV), there is an overlap between two or more classes. It shows that discrimination using a single variable is not possible. We therefore considered all three variables to discriminate the snow classes. To show the potential of these variables together in discriminating a snow class, principal component factor analysis (PCA) is used (Davis 1986; Wilkinson et al., 1992; Sturm et al., 1995). A varimax rotation (Sturm et al., 1995) was then used to maximize and minimize the component loading along the eigenvectors. A two factor plot based on three variables is plotted (Fig. 2), which suggested that 97% of the variance can be explained using two factors. The figure shows a separate group for faceted and depth hoar snow. This might be due to the similar mechanical properties of these classes. A particular area of concern is also shown in the dashed circle, which indicates more than two snow classes have similar properties in this region. Discriminating these classes simply from the database will require additional knowledge. The correlation matrix (Table 4) shows that MPF and SD are statistically highly correlated as also shown in Fig. 3; however they carry different physical information. The value of SD increases with increasing value of MPF for all snow classes. It was assumed in the beginning that class 1 and class 2 can be combined as new snow, but these two classes can be classified separately (Fig. 3a). Rounded grains snow shows wide scatter of SD after a penetration force of 5.0 N indicating a different class (Fig. 3b). This class could possibly be wind-packed snow consisting of highly sintered fine-grain snow. Faceted and depth hoar snow shows consistent values of SD with increasing values of MPF (Fig. 3c and d). As data of melt-freeze snow and hard melt-freeze crust are put together in the database, it suggests a further division when the MPF exceeds 5.0 N (Fig. 3e).

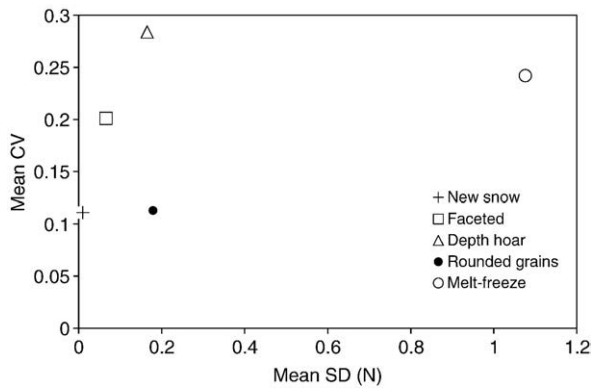
The mean values of coefficient of variation (mean CV) are also plotted against the mean values of standard deviation (mean SD) at 1 mm resolution (Fig. 4). Mean CV of new snow and rounded grains snow is found similar and mean SD of rounded grains snow and depth hoar is also found similar. This suggests that SD and CV are the characteristic values for different snow and hence, they must be used together in the snow characterization algorithm, as also supported by the PCA scores (Fig. 2). Since the variables used in the present work (MPF, SD and CV) are not too many and derived from a single variable (i.e. MPF), we envision that all three variables should be used in discriminating the snow classes.

### 2.3. Snow characterization

The snow was classified according to international snow classification (Colbeck et al., 1990, ICSSG hereafter) into the following classes: new snow as class 1 and class 2 put together, rounded grains (class 3), faceted grain (class 4), depth hoar (class 5) and melt-freeze (class 6b, 9e).

To characterize snow statistically, the values of the three variables (i.e. MPF, SD and CV) of an unknown SMP profile are first sequentially computed for 1 mm windows throughout the profile. These computed values are then compared to the database for all snow classes within an arbitrary range of  $\pm 10\%$ . The range is applied as to capture the records from the database which are close to the computed values but are not exactly the same. The program then calculates the percentage frequency of all the snow classes matched from the database and





**Fig. 4.** Mean CV is shown against mean SD for major snow classes obtained from the SMP data (Table 2). Mean SD of depth hoar and rounded grains snow is similar. Mean CV of new snow and rounded grains snow is also similar.

arranges them in descending order. The first in this order (i.e. the snow class having highest frequency) is selected as the expected snow class.

### 2.3.1. Detection of snow-air interface and snow layering

To generate a meaningful snow profile from the SMP profile, the following rules are implemented to improve the results. These rules are not in a strict sense statistical, but based on expert opinion. We believe that the expert rules can be changed with a new database collected in a different snow climate. The flow chart of the complete scheme is shown in Fig. 5.

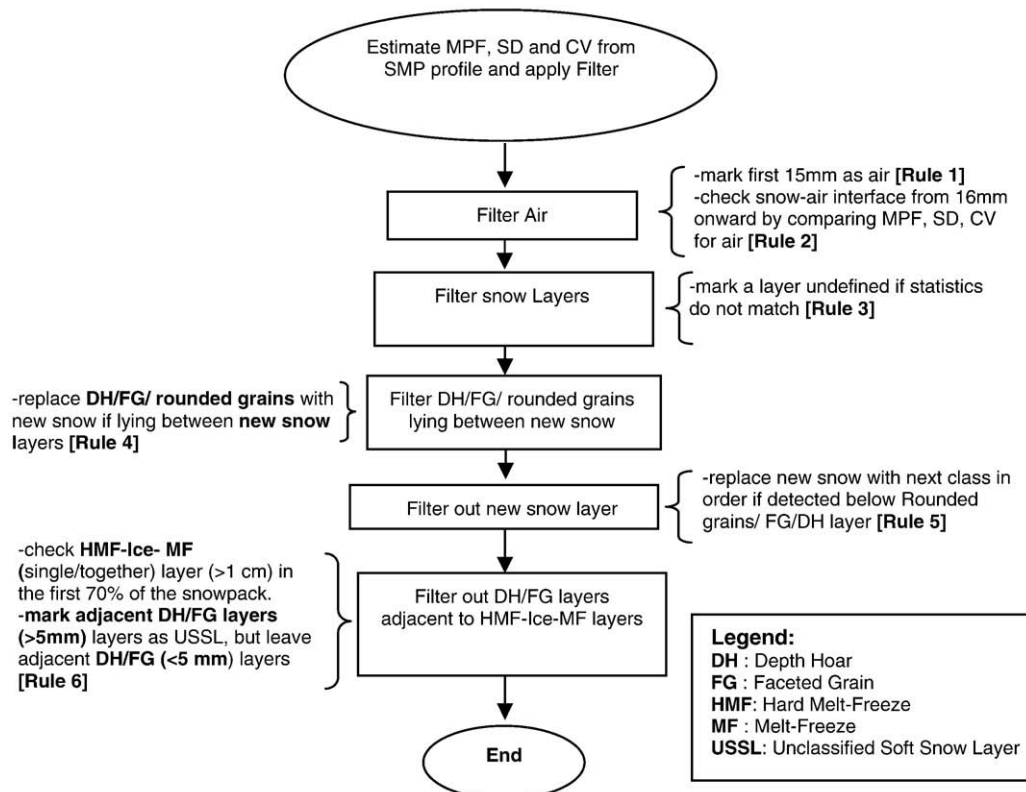
**Rule 1.** By definition, the first 15 mm of the SMP profile is marked as air. This is done because the cone tip of the SMP travels about this length in air before hitting the snow surface. But this limit can be kept

below 15 mm if necessary. The program then estimates MPF, SD and CV values averaged over 15 mm profile length.

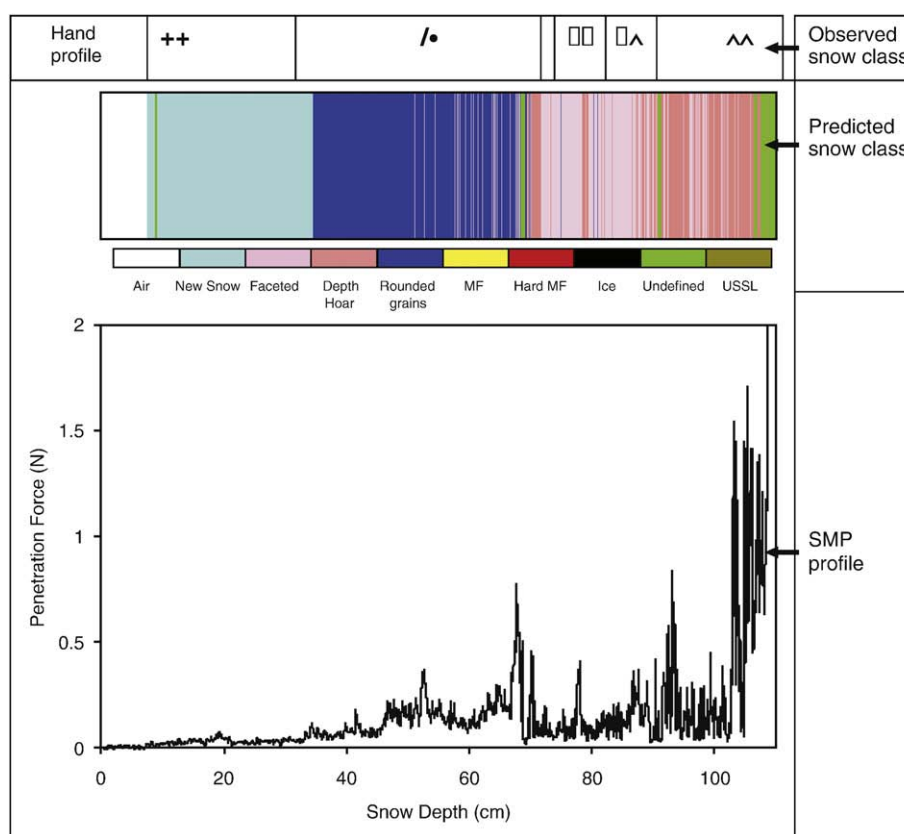
**Rule 2.** From the 15th mm onward till the end of the SMP profile, the program compares if the statistics (MPF, SD and CV) of the 16th mm correspond to a similar range as for the first 15 mm profile length. If it lies in the range  $\pm 10\%$  of the first 15 mm length, it is considered as air. The point where it first detects a non-air data (i.e. a snow-air interface) is bookmarked. From this point onwards and up to 10 mm, the search for the air records is analysed, and this is followed recursively until a particular 10 mm profile length without an air record is found. Once the snow-air interface is marked, it is ensured that no air records appear further down in the SMP profile. In case it does, it is combined with the adjacent thicker snow layer. This way, the detection of the snow-air interface is completed.

**Rule 3.** Once the snow-air interface is fixed, different snow layers are classified according to MPF, SD and CV. If statistics of a snow layer do not match with any records from the database, it is marked as undefined snow class and retained as it is. This situation is possible if SMP profile is taken from an area having large variations in the snow properties. However, in the present work this will be uncommon since all SMP profiles are compared for snowpack near Davos and statistics are allowed to vary within a range of  $\pm 10\%$ .

**Rule 4.** Once snow layers are examined according to rule 3, new snow layers are examined in the uppermost snow layers. This rule ensures that no faceted grain, depth hoar and/or rounded grains snow layers should occur between new snow layers. If the condition occurs, the layer is also marked as new snow layer. It is unlikely that two adjacent new snow layers would allow growing a faceted grain, depth hoar or rounded grains snow layer between them. However, there are several cases where near surface snow layer may grow into faceted layer due to radiation re-crystallization (Armstrong, 1985), near surface faceting (Colbeck, 1982; Fukuzawa and Akitaya, 1993; Birkeland, 1998; Birkeland et al., 1998; Fierz, 1998; Hardy et al., 2001) and



**Fig. 5.** Flowchart for snow class identification scheme.



**Fig. 6.** Snow layering of a SMP profile (sani0002) has been reproduced. This profile is taken on 02nd Feb 2005 near Davos (Table 1). More than 82% of its records matched those in the database. The probability of snow classification after comparison is estimated as 0.8.

surface hoar growth (Colbeck, 1988; Colbeck, 1989) and thereafter superimposed by new snow. The present algorithm lacks this information due to the limited database and therefore it will not predict a faceted layer if it is actually present.

**Rule 5.** This rule does not permit a new snow layer below rounded grains, faceted grains or depth hoar snow layer. In the present algorithm, it is assumed that snow layers like rounded grains, faceted grain or depth hoar are older layers (metamorphosed snow) and can generally not be found above a new snow layer. If a new snow layer is detected below metamorphosed snow, the new snow is replaced by the next snow class having the second highest frequency in order. It is possible that a new snow layer can be covered by a layer formed of rounded grains, if for example the wind picks up progressively at the end of a snowfall. Non mobilized snow that fell at the start of the precipitation can be preserved for days under newly formed windpack. The limitation of this rule is due to insufficient data.

**Rule 6.** In addition to major snow classes discussed before, a specific snow class is introduced, which is mechanically weak and defined as “unclassified soft snow layer” (USSL). USSL is introduced because it was assumed that a thin new snow layer (<5 mm) adjacent to a hard layer (melt-freeze/ice) has possibly converted into faceted grains due to radiation recrystallization, melt-layer recrystallization, and diurnal recrystallization (Colbeck, 1982; Armstrong, 1985; Fukuzawa and Akitaya, 1993; Fierz, 1998; Birkeland, 1998; Birkeland et al., 1998; Colbeck and Jamieson, 2001; Hardy et al., 2001; Jamieson and Johnston, 2001; Greene and Johnson, 2002). This layer is retained in the classified profile and not allowed to change by any rules. This layer can also expected to be a layer of wet snow or a new snow layer (other than faceted and depth hoar snow layer) if it is thicker than 5 mm. Since the criterion of 5 mm is arbitrary, the precise characterization of this layer is not

possible and the layer is retained as USSL. This completes the rule application process.

#### 2.4. Probability of snow characterization

To quantify the accuracy of the algorithm, the estimated snow class is cross-checked from the hand profile. The ratio of the correctly estimated snow layer thickness to the total snow thickness for a

**Table 5**

Estimated snow class using present algorithm is shown for three SMP control profiles

| Name of SMP profiles | Snow class observed during hand profile | Snow class estimated (%) |        |        |    |        |      |  |
|----------------------|-----------------------------------------|--------------------------|--------|--------|----|--------|------|--|
|                      |                                         | 1, 2                     | 3a, 3b | 4a, 4b | 5a | 6b, 9e | USSL |  |
| 040102-5WJ-00230005  | New snow (1,2)                          | 100                      | 0      | 0      | 0  | 0      | 0    |  |
|                      | Rounded grains (3a, 3b)                 | 0                        | 40     | 50     | 10 | 0      | 0    |  |
|                      | Faceted (4a, 4b)                        | 0                        | 0      | 75     | 22 | 3      | 0    |  |
|                      | Depth hoar (5a)                         | 0                        | 40     | 0      | 60 | 0      | 0    |  |
|                      | Melt-freeze (6b, 9e)                    | 0                        | 25     | 0      | 25 | 50     | 0    |  |
| 040331-5WJ-00550035  | New snow (1,2)                          | 75                       | 0      | 25     | 0  | 0      | 0    |  |
|                      | Rounded grains (3a, 3b)                 | 0                        | 100    | 0      | 0  | 0      | 0    |  |
|                      | Faceted (4a, 4b)                        | 0                        | 0      | 75     | 0  | 0      | 25   |  |
|                      | Depth hoar (5a)                         | 0                        | 25     | 0      | 50 | 25     | 0    |  |
|                      | Melt-freeze (6b, 9e)                    | 0                        | 0      | 0      | 0  | 100    | 0    |  |
| 031129-5WJ-00190015  | New snow (1,2)                          | 100                      | 0      | 0      | 0  | 0      | 0    |  |
|                      | Rounded grains (3a, 3b)                 | 0                        | 70     | 15     | 15 | 0      | 0    |  |
|                      | Faceted (4a, 4b)                        | 0                        | 0      | 45     | 45 | 10     | 0    |  |
|                      | Depth hoar (5a)                         | 0                        | 0      | 40     | 60 | 0      | 0    |  |
|                      | Melt-freeze (6b, 9e)                    | 0                        | 0      | 0      | 0  | 100    | 0    |  |

The Snow class code according to the International Classification (Colbeck et al., 1990) is shown in the parentheses. The hatched cells show the percentage of correctly predicted snow class vis-à-vis the actual snow class. The comparison of one SMP profile 031129\_5WJ\_00190015 is demonstrated in Fig. 7.

known class gives the probability ( $p(A)$ ), which is defined as (Lipschutz, 1981):

$$p(A) = \frac{X}{Y} \quad (1)$$

It shows that a snow class  $A$  of total thickness  $Y$  in a SMP profile can be classified correctly for a thickness of  $X$ . If  $X=Y$ , the probability of a snow class characterization will be 1. This way the probability is calculated for each snow class.

### 3. Results

The method (2.3) was first tested for the training dataset to ascertain the capability of this scheme. This was done for a SMP profile (sani0002) taken on 02 Feb 2005 (Table 1), which already had sufficient records (82%) in the database. Fig. 6 shows a classified snow profile from this SMP profile. The hand profile is also shown above the estimated profile for comparison. More than 80% profile length of the predicted snow classes matches with the hand profile giving a probability of 0.8, which is acceptable.

To validate the present method, three SMP profiles (040102\_5WJ\_00230005, 040331\_5WJ\_00550035 and 031129\_5WJ\_00190015) taken at the SLF-study plot Weissfluhjoch were used as testing dataset. Both, hand profiles and SMP profiles were compared from top down for

estimated snow classes. The snow-air interface was selected as top reference layer and a prominent hard or soft snow was selected as bottom reference layer in both the profiles (hand and SMP profile). Once the reference layers are known, the SMP profile is adjusted along the snow depth to coincide these two reference heights for comparison. Table 5 shows the results of these control SMP profiles classified using the present algorithm. The control SMP profiles are independent from the profiles considered in the classification database. It shows that many snow layers are characterized with more than one snow class. A particular thickness of rounded grains snow was matched by 40% of rounded grains records but the remaining length of this layer was characterized by faceted and depth hoar (2nd row, Table 5), and therefore characterizing such a layer with a probability of 0.4 is achieved.

An automated snow profile generated from the SMP profile (031129\_5WJ\_00190015) is shown without implementing expert rules (Fig. 7) and with the implementation of the expert rules (Fig. 8). Most of the estimated snow classes have been found matching with the observed snow class. Some of the faceted and unclassified layers in Fig. 7 are changed to USSL (through rule 6) as shown in Fig. 8. A hard MF layer that is picked up by the scheme is also shown because of the higher value of penetration resistance assigned to this layer.

Finally, results from Table 5 are used to calculate the accuracy of the present approach in characterizing different snow classes. The combined results from Table 5 are shown in Fig. 9. The probability of

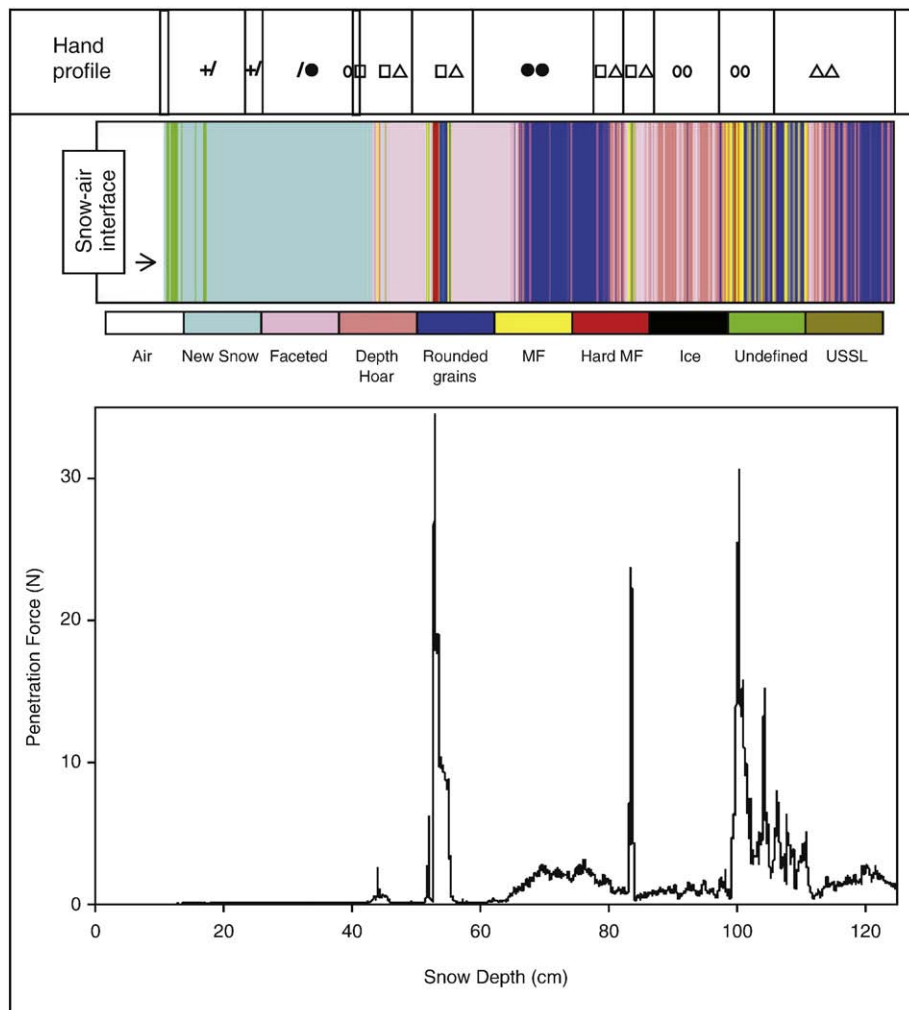


Fig. 7. The snow layering of SMP profile (031129\_5WJ\_00190015) is reproduced. Hand profile data is also shown. Snow characterization is shown without applying expert rules.

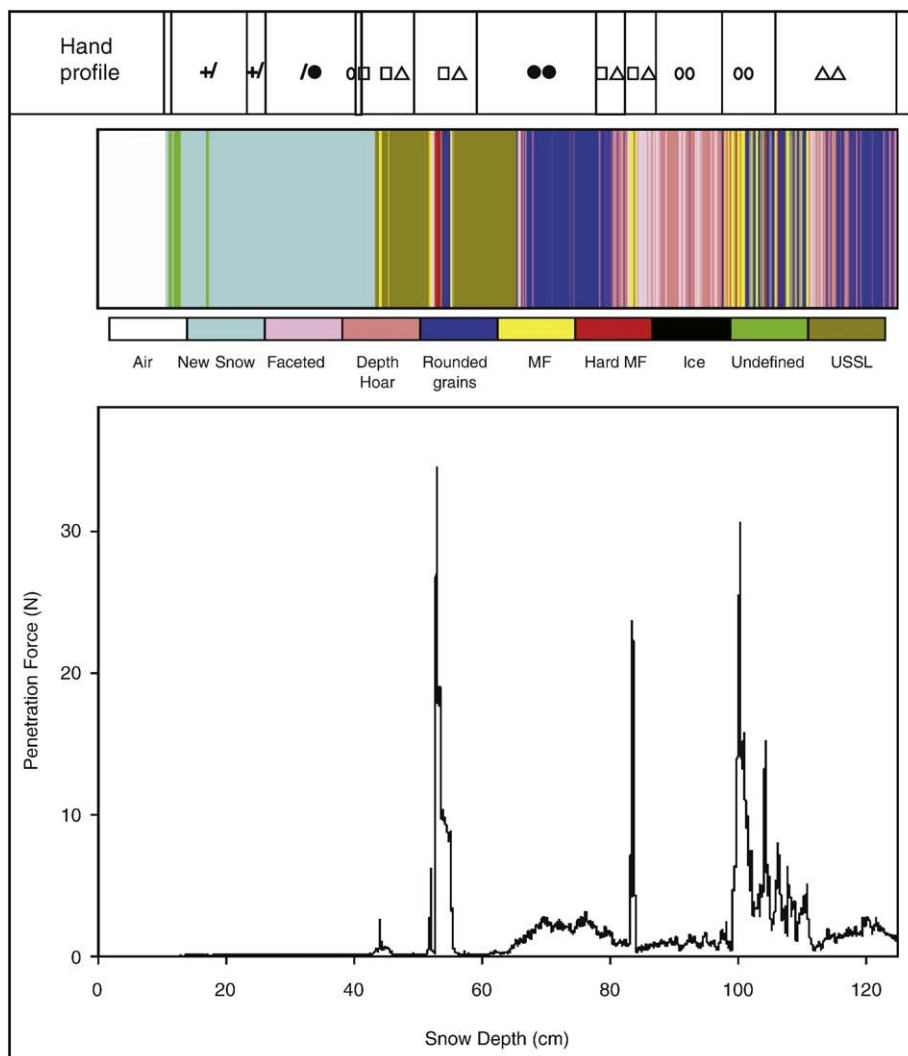


Fig. 8. Snow characterization of SMP profile (031129\_5WJ\_00190015) after the rule application is shown. Notice that some faceted snow layers in Fig. 7 are converted into unclassified soft snow layer (USSL).

snow class characterization was estimated lowest (0.68) for faceted snow and highest (0.94) for new snow. Rounded grains snow, depth hoar and melt-freeze snow were characterized correctly with a probability of 0.85, 0.75 and 0.84 respectively.

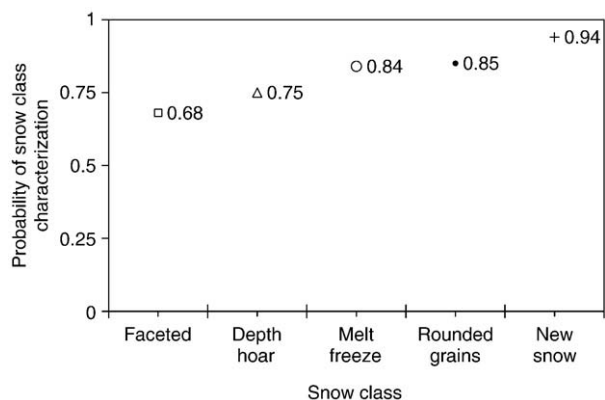


Fig. 9. Probability of snow class prediction for five major snow classes is shown as computed from Table 5. New snow (class 1 and class 2) is characterized well among all the now classes with a probability of 0.94 and faceted snow is classified the least with a probability of 0.68.

#### 4. Discussion

An observer-independent snow classification is possible using the high-resolution (4  $\mu\text{m}$ ) penetration force data of the SnowMicroPen. More than one snow class (sub classes) can be identified for a known snow class (Table 5). This suggested that a snow class can inherit the properties of other snow classes due to transition from one form to another. In the present classification scheme, the computed values of MPF, SD and CV from SMP profiles were allowed to vary arbitrary within a limit of  $\pm 10\%$ . This was done because some snow classes had only few records, such as melt-freeze and depth hoar with only 4.4% and 8.2% respectively. The limit of  $\pm 10\%$  can be reduced and finally removed when a larger database is available.

Finally, a scheme is employed where SMP profile data is used to classify five major snow classes and generate an automatic snow profile. Fresh snow (class 1) and decomposing and fragmented precipitation particles snow (class 2) were put together in the database because of their similarity and therefore retained under the new snow class. Presently, we could not distinguish between class 1 and class 2 because of the limited database. The PCA score also does not show much promise on this aspect. It would also be useful to differentiate rounded grains snow formed by equilibrium metamorphism under windless conditions from wind-packed snow. The data analysed for rounded grains snow generally forms a single cluster with small



deviation towards melt-freeze snow (Fig. 2). Melt-freeze snow (class 6b) and hard melt-freeze crust (class 9e) have been put together due to their high strength and structural similarity. However snow forming due to small numbers of melt-freeze cycles should be distinguishable from large numbers of melt-freeze cycles.

The characterization of faceted snow was difficult among all snow classes. It was expected that characterization should be better for faceted snow than for any other snow classes due to highest records in the database. The reason for this divergence is the characterization scheme that changes some of the faceted layers into USSL according to rule 6, which reduces the prediction level for faceted snow layers. In spite of being recognized well by visual inspection of SMP profiles, in most cases faceted and depth hoar snow layers are difficult to separate from each other as they have similar SMP characteristics (Fig. 2). The combined classification of faceted and depth hoar snow is however satisfactory, at least for SMP profile 031129\_5WJ\_00190015 (Table 5). This suggests that these classes though distinguishable physically, respond similar in mechanical sense against SMP penetration.

In the present work, records of near surface faceting snow and surface hoar were never found in the SMP profiles and therefore these layers are not included in the scheme and the present scheme has this limitation. The impact of this limitation on estimating the snowpack stability is very decisive as often the failure layers are the result of near surface faceting (Birkeland, 1998; Birkeland et al., 1998). As the present work is a preliminary step towards quantifying snow classes, more effort is still required before such schemes are made operational. As we have not carried out the multivariate analysis in present work, it is proposed that a discriminant function be derived for snow classification once sufficient data is available.

As explained earlier, there may be situations wherein a snow layer can contain more than one snow class due to transitions from one class to another, and their origin may be different (e.g. melt-freeze layer originating from depth hoar), we found that the traditional classification is not very precise about this. The present scheme can pick such snow class having second highest frequency matched in the database (Table 5). This way, a snow class and its sub-class can be characterized for a given snow layer. The method developed here is tuned such that the SMP profile corresponded as much as possible to the representation in hand snow profiles. This has the advantage that the complex information in a snow profile can be used in a structured form to extract snowpack stability and to follow temporal evolution. Although the present classification algorithm is modeled after the traditional snow classification, many more layers are found. This could be considered as weakness in the automated scheme. However, in our opinion, this observation is the result of the implicit generalization, which is often done by the hand snow profile description, when practitioners focus on layers which are critical for stability. We think that such generalizations in hand profiles are detrimental if the profiles are used to validate numerical snowpack simulations, e.g. CROCUS (Brun et al., 1992) and SNOWPACK (Bartelt and Lehning, 2002).

The characterization of snow and extraction of snow class information using a penetrometer can also be used to infer specific surface area (SSA) and optically equivalent grain size (OED), respectively and correlation lengths (Taillandier et al., 2006; Nolin and Dozier, 2000; Mätzler, 2002). Kerbrat et al. (2008) could show that a range of SSA is closely linked to a certain snow type. Specific surface area and optically equivalent diameter are related by a simple geometrical relation (Matz and Schneebeli, 2006). This opens the possibility to use the snow class information derived from the SMP to infer the Snow Area Index (Taillandier et al., 2006), to measure optically equivalent grain size derived from hyperspectral imaging (Nolin and Dozier, 2000), and to evaluate the correlation length of snow, as used e.g. in modeling microwave emission. The disadvantage of an empirical correlation to SSA and OED is potentially offset by the advantage that many profiles can be measured very rapidly, and evaluated. This may be especially useful for microwave modeling, as

the spatial variability of the snowpack is often very large and an optimal sampling scheme very difficult to develop without a-priori knowledge of distribution of the properties of the snow profiles, which is usually the case. A similar reasoning is true for the measurement of the Snow Area Index, where exhaustive spatial sampling would be often more important for an unbiased estimate than the absolute precision of a single measurement.

## 5. Conclusion

The present algorithm demonstrates the capability of high-resolution micro-penetrometer data for extracting stratigraphic information correlated to traditional visual snow classification for Alpine snow. The strength of the high-resolution penetrometer is presented with a view to interpret the snow classes from the SMP profile. It is shown that standard deviation and coefficient of variation together with mean penetration force at 1 mm resolution can characterize major snow classes. It was challenging to characterize snow classes that have similar values of standard deviation and coefficient of variation. Knowledge based rules applied in the scheme helped in identifying such snow classes and generating a snow profile. Finally, the estimated SMP profiles are validated using the hand profile data. The probability of snow class characterization was estimated as 0.94 (highest) for new snow, 0.85 for rounded grains snow, 0.84 for melt-freeze snow, 0.75 for depth hoar and 0.68 (lowest) for faceted snow. This concept is by itself difficult, as the traditional classification of a snow profile inherits some observer-dependent bias. The advantage of the grain shape classification may be helpful for future snowpack stability estimation since grain shape information is indicative of trends in bonding. Converting the SMP signals to a traditional snow profile would be a helpful training tool for forecasters making the transition to interpreting SMP profiles. The present work provides a new concept of an observer-independent method of snowpack analysis. A primary drawback of numerical simulation models, such as CROCUS and SNOWPACK, is that the algorithms to track the evolution of snow structure are difficult to validate. Human observations are simply inadequate for this task. As such, numerical avalanche forecasting, which would be extremely valuable in regions where frequent human observations are not possible, can simply not be advanced. We believe the algorithm presented in this paper, combining objective SMP measurements with knowledge-based rules, could be extremely helpful in validating numerical models and thus opening the door to new model based forecasting techniques.

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